

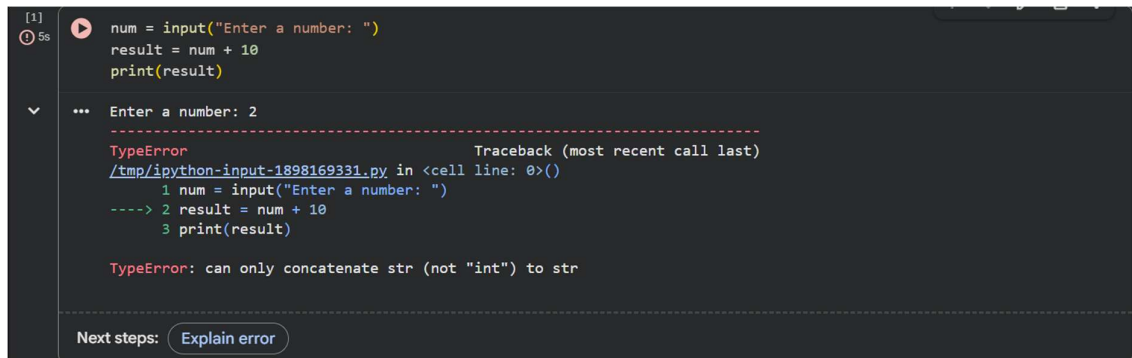
School of Computer Science and Artificial Intelligence

Lab Assignment # 7.2

Program : B. Tech (CSE)
Specialization : -
Course Title : AI Assisted Coding
Course Code : 23CS002PC304
Semester : II
Academic Session : 2025-2026
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Enrollment No. : 2403A51L04
Batch No. : 51
Date : 30/01/26

Submission Starts here**Screenshots:****Task 1 – Runtime Error Due to Invalid Input Type****(Buggy Code):**

```
num = input("Enter a number: ")
result = num + 10
print(result)
```

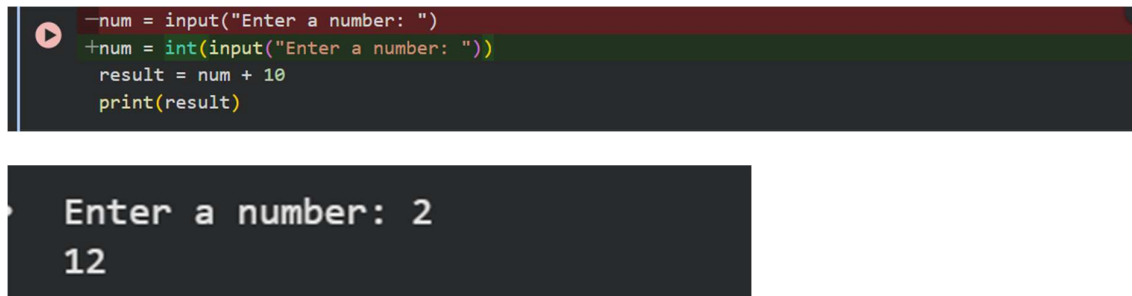


```
[1] 5s ▶ num = input("Enter a number: ")
      result = num + 10
      print(result)

*** Enter a number: 2
-----
TypeError                                Traceback (most recent call last)
/tmp/ipython-input-1898169331.py in <cell line: 0>()
      1 num = input("Enter a number: ")
----> 2 result = num + 10
      3 print(result)

TypeError: can only concatenate str (not "int") to str

Next steps: Explain error
```

Output:

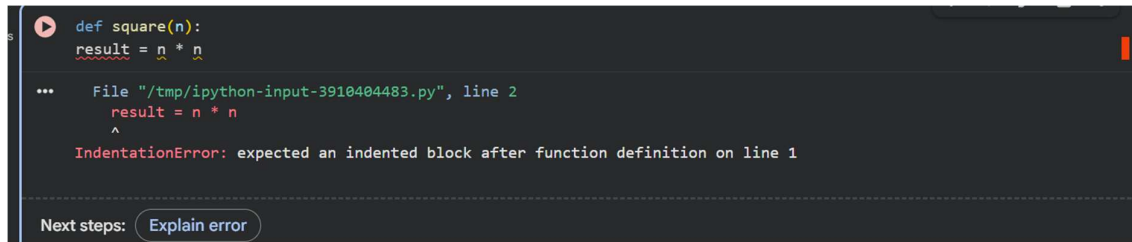
```
▶ -num = input("Enter a number: ")
  +num = int(input("Enter a number: "))
    result = num + 10
    print(result)

Enter a number: 2
12
```

Task 2 – Incorrect Function Return Value

(Buggy Code):

```
def square(n):
result = n * n
```

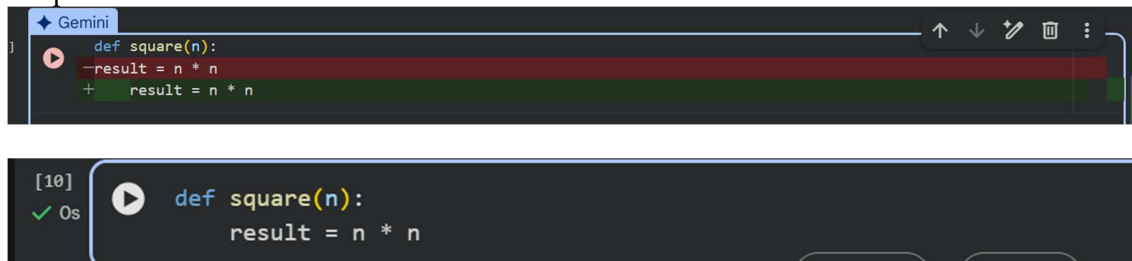


```
def square(n):
result = n * n

... File "/tmp/ipython-input-3910404483.py", line 2
      result = n * n
      ^
IndentationError: expected an indented block after function definition on line 1

Next steps: Explain error
```

Output:



```
def square(n):
    result = n * n

[10] ✓ 0s def square(n):
        result = n * n
```

Task 3 – IndexError in List Traversal

(Buggy Code):

```
numbers = [10, 20, 30]
for i in range(0, len(numbers)+1):
print(numbers[i])
```

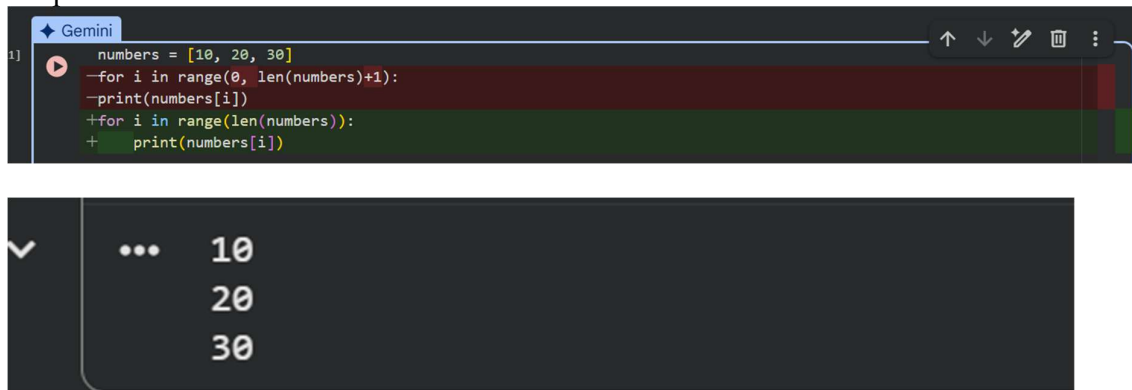


```
numbers = [10, 20, 30]
for i in range(0, len(numbers)+1):
print(numbers[i])

... File "/tmp/ipython-input-726334973.py", line 3
print(numbers[i])
      ^
IndexError: list index out of range

Next steps: Explain error
```

Output:



```
numbers = [10, 20, 30]
for i in range(0, len(numbers)):
print(numbers[i])

... 10
    20
    30
```

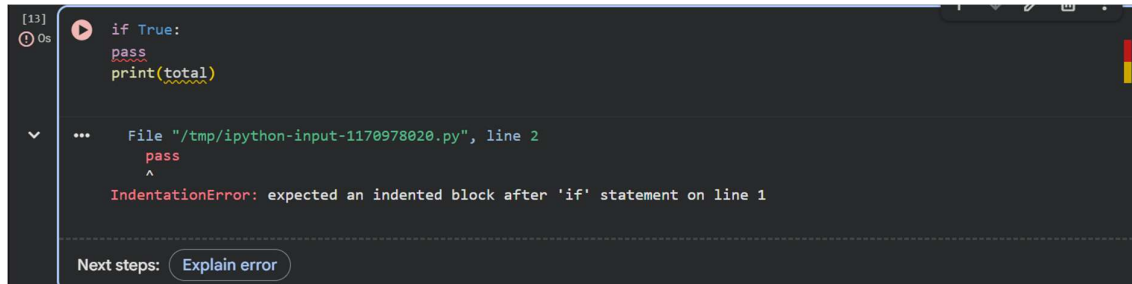
Task 4 – Uninitialized Variable Usage

(Buggy Code):

```
if True:
```

```
    pass
```

```
print(total)
```



The screenshot shows a Jupyter Notebook cell with the following code:

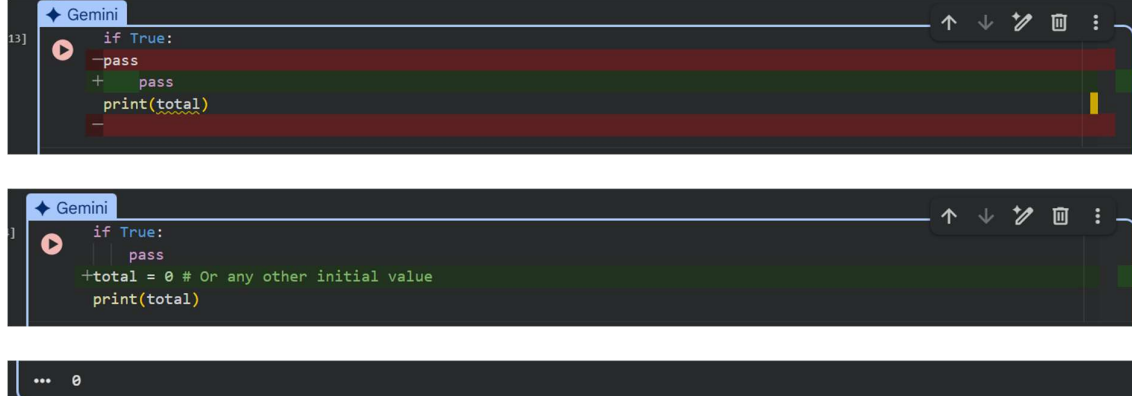
```
[13] In [13]: if True:
            pass
            print(total)
```

The code is highlighted in yellow. Below the code, an error message is displayed:

```
... File "/tmp/ipython-input-1170978020.py", line 2
      pass
      ^
IndentationError: expected an indented block after 'if' statement on line 1
```

At the bottom of the cell, there is a button labeled "Next steps: Explain error".

Output:



The first screenshot shows the code from the previous block with a red background, indicating an error. The code is:

```
[13] In [13]: if True:
            -pass
            + pass
            print(total)
```

The second screenshot shows the code with a correction: a line is added to initialize the variable 'total' before it is used in the print statement.

```
[13] In [13]: if True:
            pass
            +total = 0 # Or any other initial value
            print(total)
```

Below the code, there is a button labeled "Next steps: Explain error".

Task 5 – Logical Error in Student Grading System

(Buggy Code):

```
marks = 85
```

```
if marks >= 90:
```

```
    grade = "A"
```

```
elif marks >= 80:
```

```
    grade = "C"
```

```
else:
```

```
    grade = "B"
```

```
print(grade)
```

```
[16] marks = 85
      if marks >= 90:
        grade = "A"
      elif marks >= 80:
        grade = "C"
      else:
        grade = "B"
      print(grade)]
```

... File "/tmp/ipython-input-2691675298.py", line 3
 grade = "A"
 ^
IndentationError: expected an indented block after 'if' statement on line 2

Next steps: [Explain error](#)

Output:

Gemini

```
marks = 85
if marks >= 90:
- grade = "A"
+ grade = "A"
elif marks >= 80:
- grade = "C"
+ grade = "C"
else:
- grade = "B"
+ grade = "B"
print(grade)
```

... C